Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

764
2 0.26%
Sep 19th, 2022

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

854
-8.00 -0.93%
Sep 19th, 2022

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

706
0 0.00%
Sep 19th, 2022

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

103.10
-0.75 -0.72%
Sep 19th, 2022

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

110.70
-0.80 -0.72%
Sep 19th, 2022

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

875
5 0.57%
Week Ending Sep 16th, 2022

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2301 (Jan) RMB/t (3pm close)

705.50
-9.50 -1.33%
Sep 19th, 2022 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
October 22 USD/dmt

96.95
-0.95 -0.97%
Sep 19th, 2022 (5.30 pm Print)

SHFE Rebar
RB2301 (Jan) RMB/t

3696
10 0.27%
Sep 19th, 2022 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

21.92
-0.39 -1.74%
Sep 16th, 2022

C5, W. Australia - Qingdao USD/t

8.85
-0.94 -9.57%
Sep 16th, 2022

Steel Price

Steel Rebar (China Domestic) RMB/t

3950
-60 -1.50%
Week Ending Sep 16th, 2022

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

133.24
0.36 0.27%
Week Ending Sep 16th, 2022

Steel Inventory in China
million tonnes

12.52
0.20 1.61%
Week Ending Sep 16th, 2022

Steel Price

Steel HRC (China Domestic) RMB/t

3900
-30 -0.76%
Week Ending Sep 16th, 2022

IRON ORE PORT STOCK INDEX (IOPI)

Sep 19th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	764	2	0.3%	773	904	587	1083	102.02	0.14	0.1%	104.92	128.36	84.25	159.06
IOPI58	58% Fe Fines	706	0	0.0%	686	753	454	907	95.01	-0.12	-0.1%	93.69	107.18	64.78	134.15
IOPI65	65% Fe Fines	854	-8	-0.9%	878	1064	811	1265	114.49	-1.26	-1.1%	119.63	151.98	111.38	187.25

IRON ORE SEABORNE INDEX (IOSI)

Sep 19th, 2022		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	103.10	-0.75	-0.72%	107.70	131.45	93.75	159.45
IOSI65	65% Fe Fines	110.70	-0.80	-0.72%	118.25	152.70	102.05	185.82

MARKET COMMENTARY

DCE iron ore futures declined by 1.4% today, the main contract closed 705.5. Some traders were general attitude to sell. More steel mills tended to wait and see, and purchasing enthusiasm is not high. Total transactions returned to positive.PBF at Shandong port dealt 745-760 yuan/mt; decreased 5-10 yuan/mt than last week. According to SMM statistics, from September 12 to September 18, the total global iron ore shipments totaled 31.01 million tons, an increase of 18.1% from the previous month. Among them, Australia's shipments to China increased month-on-month, while Brazil's shipments to China decreased month-on-month. The total amount of Chinese iron ore arriving in Hong Kong was 23.47 million tons, an increase of 49.2% from the previous month. After a slight drop in shipments last week, overseas shipments have returned to stability. At present, the inventory of steel mills is low, and there is still a strong demand for replenishment before the National Day holiday. However, considering that the eve of the important national meeting in October will gradually limit production and logistics due to environmental protection factors, the room for demand growth may be relatively limited. In addition, the current macro news is more, and the

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Sep 16th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	875	5	0.6%	889	1111	730	1405	113.18	0.65	0.58%	116.18	152.24	101.00	199.00

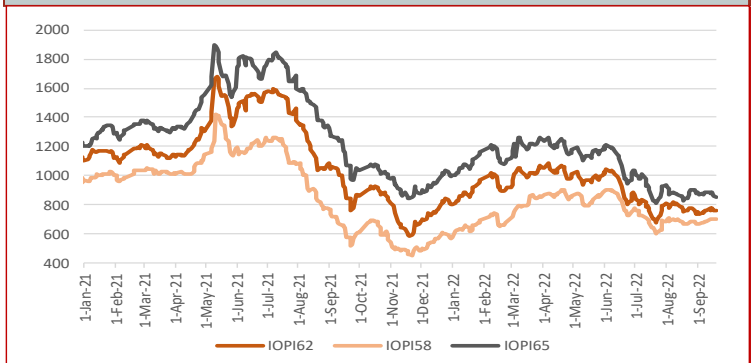
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Sep 16th, 2022				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	909	1.0%	779	1645	131.52	0.98%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	905	2.3%	780	1630	130.95	2.24%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	710	0.0%	620	1310	102.73	-0.02%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	932	1.0%	800	1752	134.85	0.96%	122.55	272.32

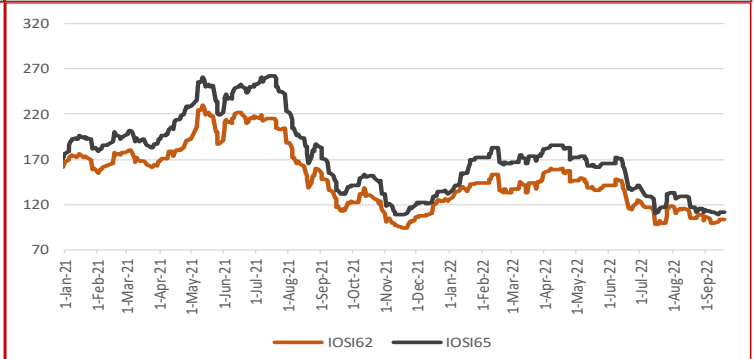
Week Ending Sep 16th, 2022	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	835.90	1.17%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.91125

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Sep 19th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	May	June	July	August	MTD	QTD	YTD	May	June	June	August	MTD	QTD	YTD
IOPI62	62% Fe Fines	981	934	770	781	773	772	904	136.86	130.13	106.01	106.64	104.92	105.33	128.36
IOPI58	58% Fe Fines	841	823	682	685	686	684	753	118.11	115.45	94.43	94.05	93.69	93.97	107.18
IOPI65	65% Fe Fines	1152	1086	915	878	878	891	1064	161.44	151.89	126.73	120.31	119.63	122.29	151.98

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Sep 19th, 2022		CFR Qingdao, USD/dry tonne						
Index	Fe Content	May	June	July	August	MTD	QTD	YTD
IOSI62	62% Fe Fines	141.17	131.96	110.07	110.64	107.70	108.58	131.45
IOSI65	65% Fe Fines	166.30	154.17	124.60	112.53	118.25	120.61	152.70

FREIGHT RATES

Sep 16th, 2022		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	8.85	-0.94	-9.57%	3.57	16.77
Tubarao - Qingdao	C3	21.92	-0.39	-1.74%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Sep 16th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	May	June	July	August	MTD	QTD	YTD	May	June	July	August	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	1232	1189	906	900	889	896	1111	164.53	159.78	120.36	118.50	116.18	117.92	152.24

IRON ORE INDEX PREMIUMS/DISCOUNTS

Sep 19th, 2022		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-58	-7.59%
IOPI65	65% Fe Fines	90	11.78%

Sep 19th, 2022		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	7.60	7.37%

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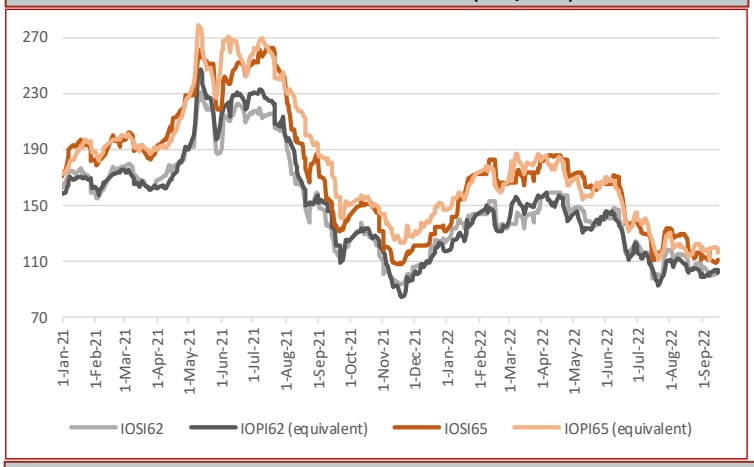
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Sep 19th, 2022	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	707	2	-57
SIMEC Fines	634	2	-130
PB Fines	734	3	-30
Newman Fines	732	2	-32
MAC Fines	714	2	-50
Jimblebar Blended Fines	628	2	-136
Carajas Fines	944	2	180
Brazilian SSF	726	2	-38
Brazilian Blend Fines	747	2	-17
RTX Fines	646	1	-118
West Pilbara Fines	676	2	-88

Sep 19th, 2022	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	98.60	-0.75	-4.50
SIMEC Fines	95.10	-0.75	-8.00
PB Fines	99.35	-0.75	-3.75
Newman Fines	102.25	-0.73	-0.85
MAC Fines	99.35	-0.75	-3.75
Jimblebar Blended Fines	91.75	-0.73	-11.35
Carajas Fines	132.65	-0.75	29.55
Brazilian SSF	106.85	-0.75	3.75
Brazilian Blend Fines	108.50	-0.73	5.40
RTX Fines	93.00	-0.73	-10.10
West Pilbara Fines	97.35	-0.75	-5.75

Sep 19th, 2022	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	706	0	0
FMG Blended Fines	715	0	9
Robe River	715	0	9
Western Fines	718	0	12
Atlas Fines	712	0	6
Yandi	699	0	-7

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	3.00	0.00
	High Grade Fe 63 - 64%	32.00	3.00
	High Grade Fe 64 - 65%	32.00	3.00
	High Grade Fe 65 - 65.5%	32.00	3.00
	Low Grade Fe	11.00	0.00
1% Alumina	High Fe Grade Al <2.25%	10.00	-4.00
	High Fe Grade Al 2.25-4%	5.00	0.00
	Low Fe Grade Al <2.25%	22.00	2.00
	Low Fe Grade Al 2.25-4%	10.00	0.00
1% Silica	High Fe Grade Si <4%	6.00	-5.00
	High Fe Grade Si 4-6.5%	21.00	0.00
	Low Fe Grade	20.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	7.00	0.00
	High Fe Grade 0.115%<P<0.15%	7.00	0.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	3.25	0.00
	High Grade Fe 63 - 64%	0.50	0.00
	High Grade Fe 64 - 65%	0.50	0.00
	High Grade Fe 65 - 65.5%	0.50	0.00
1% Alumina	High Fe Grade Al <2.25%	0.25	0.00
	High Fe Grade Al 2.25-4%	0.25	0.00
1% Silica	High Fe Grade Si <4%	2.50	0.00
	High Fe Grade Si 4 - 6.5%	1.75	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	1.50	0.00
	High Fe Grade 0.115%<P<0.15%	3.25	0.00

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-10.00	0.00	Jingtang	-5.00	0.00	Majishan	0.00	0.00	Taicang	-40.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-20.00	0.00

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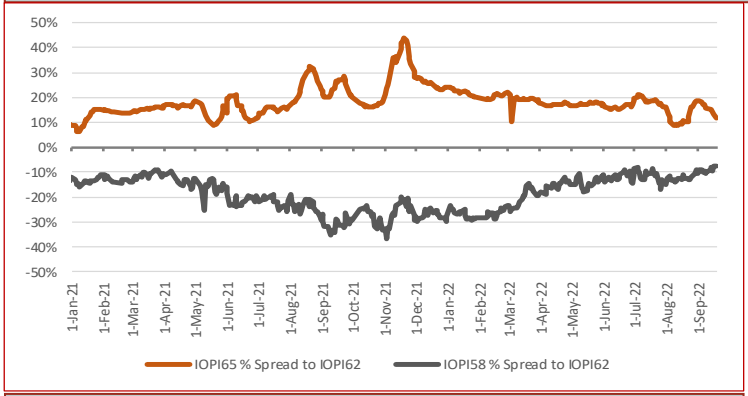
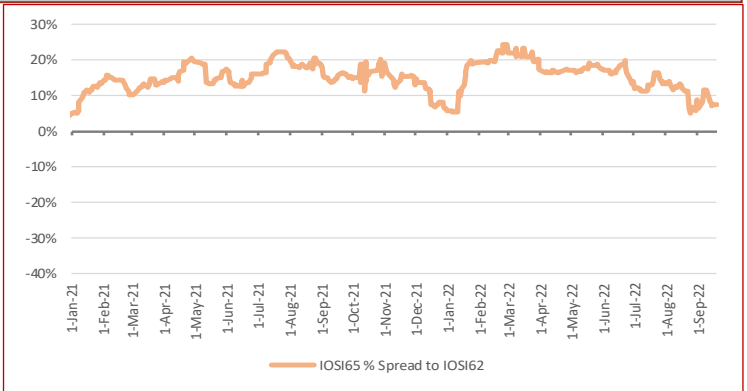
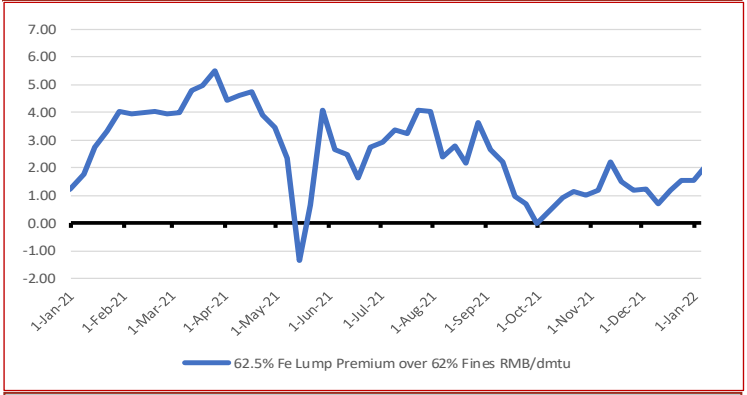
IRON ORE INDEX PREMIUMS/DISCOUNTS

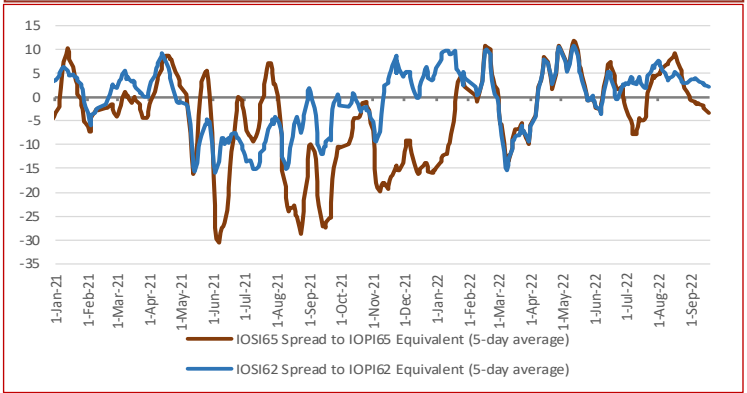




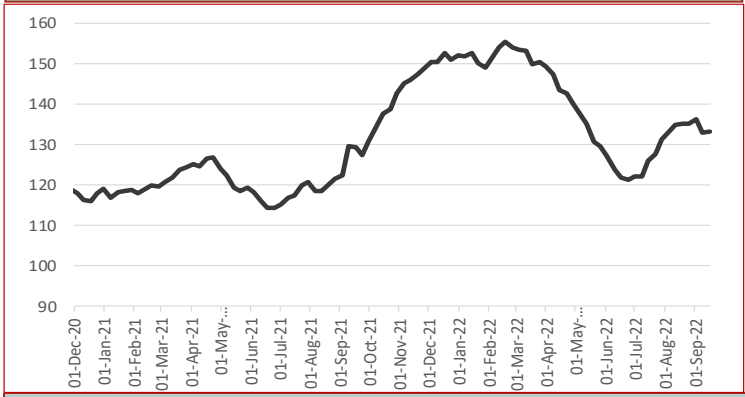
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



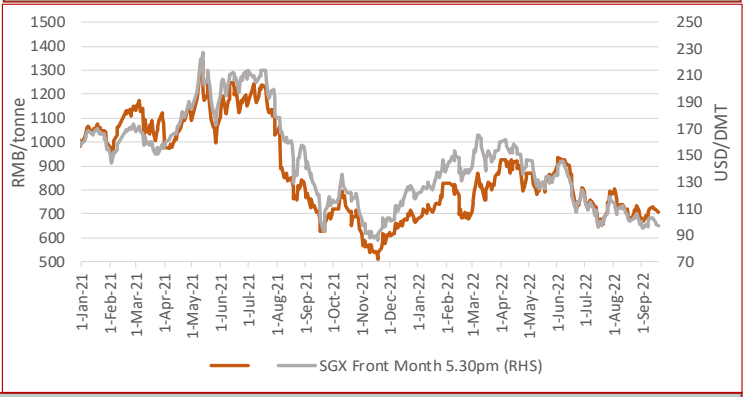
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

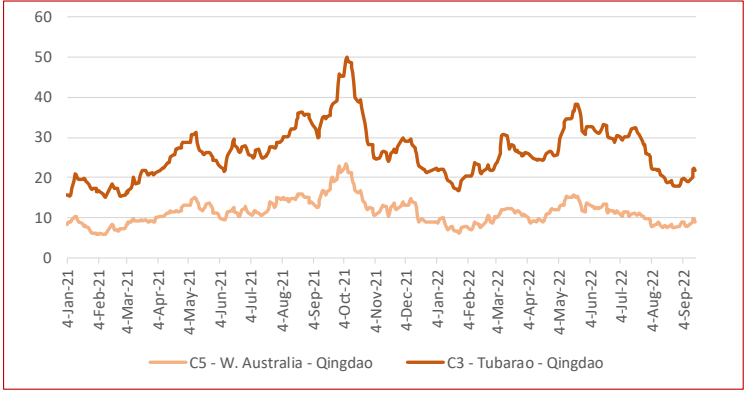
Week Ending Sep 16th, 2022

Province	This week	Change %	Low [2]	High [2]
Jingtang	12.27	-1.52%	9.84	19.20
Qingdao	24.95	1.63%	9.41	26.24
Caofeidian	12.50	2.46%	9.05	16.29
Tianjin	10.28	-2.84%	7.14	12.97
Rizhao	18.15	-1.89%	9.44	19.26
Total (35 Ports)	133.24	0.27%	98.80	155.39

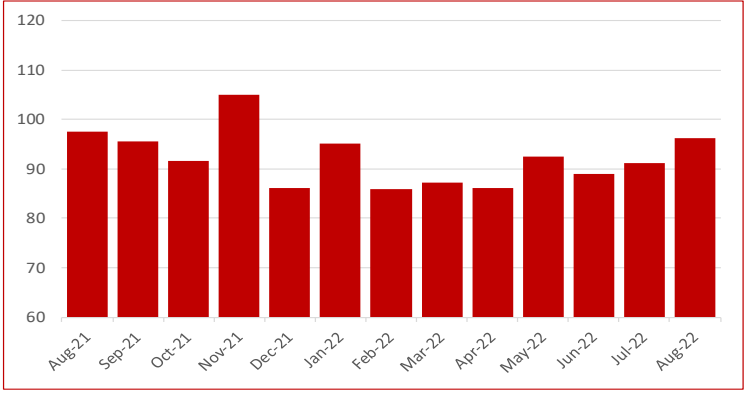
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Sep 19th, 3pm close			Sep 19th, 5:30pm		
Contract	I2301	Change	Change %	Oct. 22	Change	Change %
Closing Price	705.50	-9.50	-1.33%	96.95	-0.95	-0.97%
Vol traded ('000 lots)	70.05	-9.55	-12.00%	3.84	0.24	6.56%
Open positions ('000 lots)	76.69	0.10	0.12%	28.23	-0.52	-1.79%
Day Low	705.0	-2.50	-0.35%	96.50	-1.00	-1.03%
Day High	728.0	0.50	0.07%	100.35	-0.65	-0.64%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2022/9/16	Change	Change %
ReBar HRB400 ϕ18mm	3,950	-60	-1.50%
Wirerod Q300 ϕ6.5mm	4,210	-70	-1.64%
HRC Q235/SS400 5.5mm*1500*C	3,900	-30	-0.76%
CRC SPCC/ST12 1.0mm*1250*2500	4,350	-10	-0.23%
Medium & Heavy Plate Q235B 20mm	4,090	0	0.00%
GI ST02Z 1.0mm*1000*C	4,650	10	0.22%
Colour Coated Plate	7,550	0	0.00%

CHINESE STEEL MILL PROFITABILITY

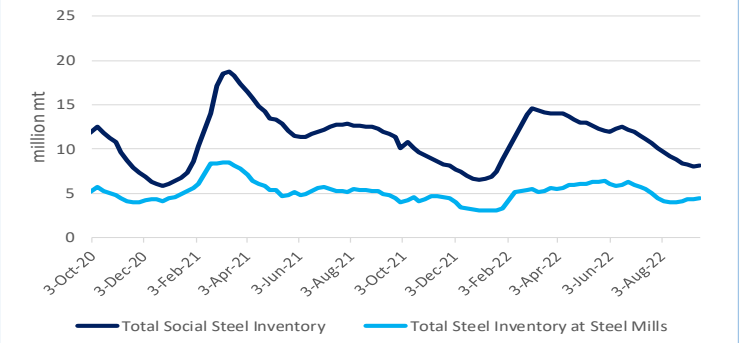
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	103.50	4	Mmi CFR Equivalent index for 1st Feb
Coke	2,910	0	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	3,140	0	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,565	20	Q234, incl. tax
Rebar cost - Blast furnace	4,062	23	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-112	-73	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	4,028	22	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	-148	-72	based on Shanghai prices, incl. tax

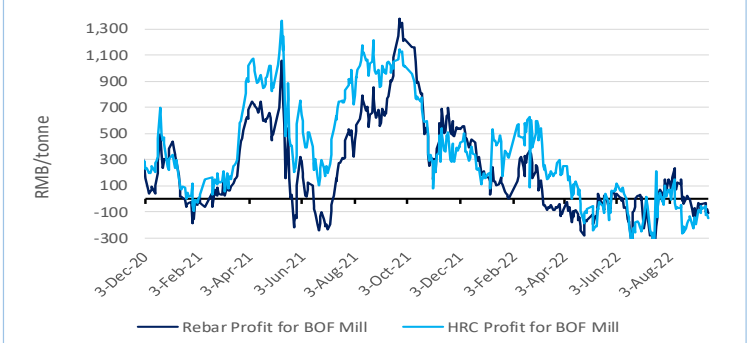
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

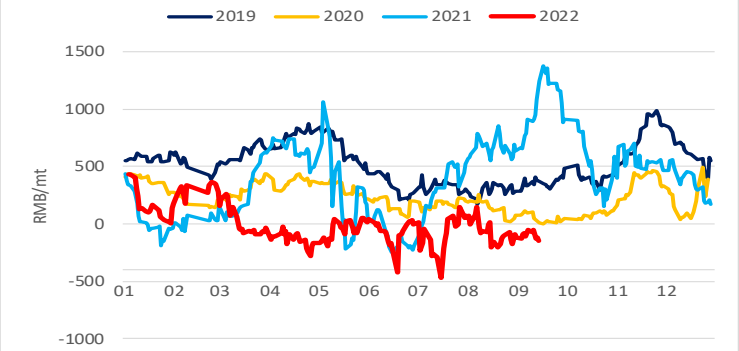
CHINESE STEEL INVENTORIES



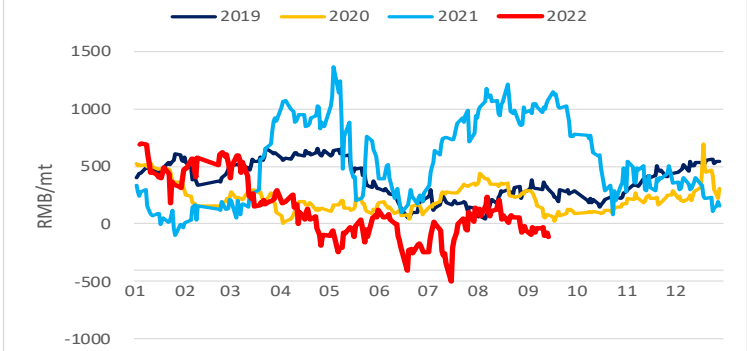
CHINESE STEEL MILL PROFITABILITY



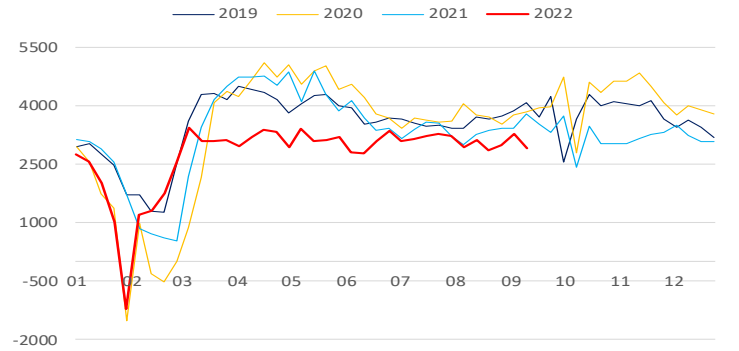
CHINESE STEEL MILL PROFITABILITY—Rebar



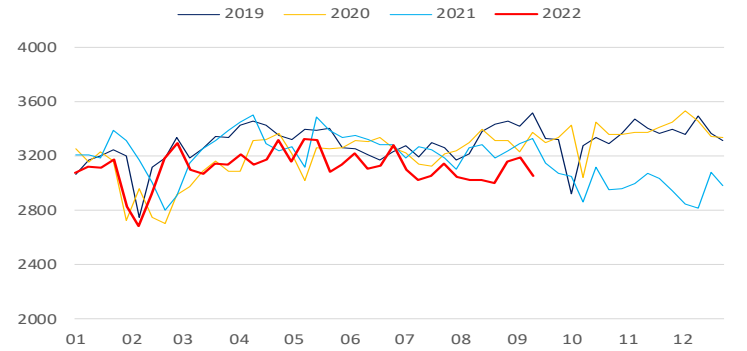
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



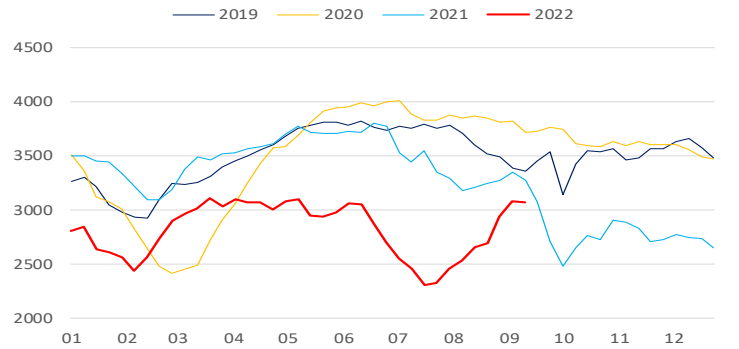
CHINESE STEEL CONSUMPTION—Rebar



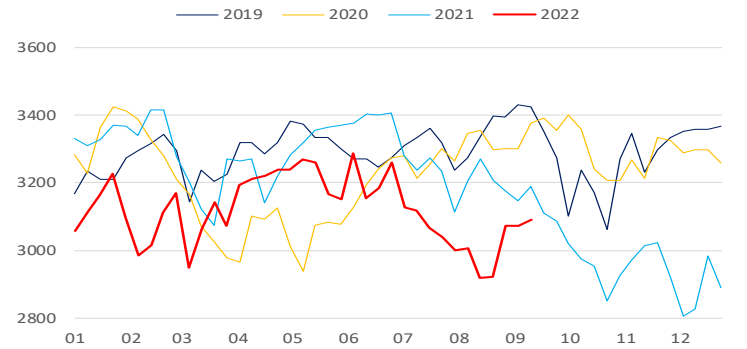
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Sep 19th, 2022	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Sep 19th, 2022	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

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